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TWO PARALLEL QUEUES CREATED BY ARRIVALS WITH TWO DEMANDS I

L. FLATTO† AND S. HAHN‡

Abstract. We analyze the double queue that arises when arriving customers simultaneously place two demands handled independently by two servers. It is assumed that the customer arrivals form a Poisson process with mean 1, the servers have exponential service times with rates α, β and $1 < \alpha \leq \beta$, which implies stability of the queue. The equations for the equilibrium probabilities $p_{ij} = P$ (i customers in α -queue, j customers in β -queue) are converted into a functional equations for $P(z, w) = \sum p_{ij} z^i w^j$, which exhibits a relation between $P(z, 0)$, $P(0, w)$ on the portion $|z|, |w| \leq 1$ of $S = \{(z, w): (1 + \alpha + \beta)zw - \alpha w - \beta z - z^2 w^2 = 0\}$. S is a Riemann surface of genus 1 which is parametrized by a pair of elliptic functions $z = z(t)$, $w = w(t)$. The functional equation for $P(z, w)$ is converted into a set of conditions on $P(z(t), 0)$, $P(0, w(t))$, which in turn lead to the determination of $P(z, w)$. From this, one obtains asymptotic formulas for p_{ij} as either i or $j \rightarrow \infty$.

1. Introduction. In this paper we analyze the double queue that arises when arriving customers simultaneously place two demands handled independently by two servers. As an application of such queues we mention the inventory control of data base systems. A common inaccuracy which arises in such systems is the deviation of the number of items recorded in the data base from the number of items present in stock. The deviation is due to the fact that the updating in the system may be made at times different from those when an item is actually added or removed from stock. If the deviation is due to removals only, then the actual and recorded number of items may be treated as two parallel queues created by the same arrival process, each having its own servicing time distribution.

We assume that the customer arrivals form a Poisson process with mean 1 and that the servers have exponential service times with rates α, β . The servers with rates α and β are referred to as the α - and β -servers, their respective queues as the α - and β -queues, and the double queue as the α, β -queue.

Let $X_1(t)$, $X_2(t)$ be the respective number of customers waiting or in service at time t in the α - and β -queues. $X(t) = (X_1(t), X_2(t))$ is a Markov chain whose state space consists of the integer pairs (i, j) , $i, j \geq 0$. Intuitively, this chain is expected to be stable iff $\alpha, \beta > 1$. That is, if $p_{ij}(t) = P(X_1(t) = i, X_2(t) = j)$, then $\lim_{t \rightarrow \infty} p_{ij}(t) = p_{ij} > 0$, $0 \leq i, j < \infty$, where the p_{ij} 's are independent of the initial distribution $p_{ij}(0)$ and $\sum p_{ij} = 1$. The numbers p_{ij} are called the equilibrium probabilities. A proof of stability is given in § 2. The condition $\alpha, \beta > 1$ is imposed from now on and we address ourselves to the problem of computing p_{ij} , which reduces to the problem of computing the generating function $P(z, w) = \sum p_{ij} z^i w^j$. Since $\sum p_{ij} = 1$ and $p_{ij} > 0 \forall i, j$, the series converges for $|z|, |w| \leq 1$.

In § 2, we write down the equilibrium equations for p_{ij} and convert these into the following equation for $P(z, w)$

$$(1.1) \quad Q(z, w)P(z, w) = N(z, w), \quad |z|, |w| \leq 1,$$

where

$$(1.2) \quad \begin{aligned} Q(z, w) &= (1 + \alpha + \beta)zw - \alpha w - \beta z - z^2 w^2, \\ N(z, w) &= \beta z(w - 1)P(z, 0) + \alpha w(z - 1)P(0, w). \end{aligned}$$

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It follows from (1.1) that $N(z, w) = 0$ on the portion $|z|, |w| \leq 1$ of the algebraic curve S defined by $Q(z, w) = 0$. The algebraic curve, or equivalently, the Riemann surface S , is of genus 1, i.e., a torus. In § 3, we parametrize S by a pair of elliptic functions $z = z(t)$, $w = w(t)$; t is referred to as a uniformizing parameter on S . We show, in § 5, that $P(z(t), 0)$, $P(0, w(t))$ can be continued as meromorphic functions to the entire t -plane and we convert the functional equations for $P(z, w)$ into a set of conditions on $P(z(t), 0)$, $P(0, w(t))$ (Theorem 5.1). In § 6, these conditions are employed to determine $P(z, 0)$, $P(0, w)$ and, hence, $P(z, w)$.

The idea of introducing a uniformizing parameter to solve functional equations similar to (1.1) occurs in the work of Malyshev [5] on random walks on positive two-dimensional lattices. The method is also employed in the paper of Flatto and McKean [3] on parallel queues. In [3] the surface S to be parametrized is of genus 0, a sphere. In [5] S is of genus 1, a torus. Thus the functional equation (1.1) is a special case of the one treated in [5]. Here we take this special case further, and use standard facts about elliptic functions to identify $P(z, 0)$, $P(0, w)$ as certain algebraic functions of z, w .

For an alternative approach to our problem, where the study of the functional equation for $P(z, w)$ is reduced to that of a boundary value problem for either $P(z, 0)$ or $P(0, w)$, the reader is referred to the papers of Fayolle and Iasnogorodski [2] and Foschini [4].

It is instructive to compare the results of the present paper and those in [3] which emphasize the role of the genus of S . In [3], $P(z, 0)$ and $P(0, w)$ are extended to be single-valued meromorphic functions on the entire z - and w -planes. In the present work, $P(z, 0)$ and $P(0, w)$ are extended to be algebraic functions which are not single-valued on the z - and w -planes. The reason for this difficulty stems from the fact that, as opposed to the genus 0 case, the mapping $\phi(t) = (z(t), w(t))$ from the t -plane to the surface S is not one-to-one. Thus, while the functions $P(z(t), 0)$, $P(0, w(t))$ are single-valued on the t -plane, this is not the case for the functions $P(z, 0)$, $P(0, w)$ on S . In [3], we obtain a partial fraction decomposition for $P(z, w)$ which leads to exact and asymptotic formulas for p_{ij} which are uniform in i, j . This decomposition is not available for multiple-valued functions and the exact formulas are no longer present. We do, however, derive, in § 7 asymptotic formulas for p_{ij} from the expressions for $P(z, 0)$, $P(0, w)$. The asymptotics are less complete than those of [3] as they fail to be uniform in i, j . In a subsequent paper, we shall derive uniform asymptotics for p_{ij} and use these to study the interdependence of the two queues.

2. The equilibrium equations. The functions $p_{ij}(t)$ satisfy the Kolmogorov forward equations [1, p. 48]. The equilibrium equations are obtained from these by setting $dp_{ij}/dt = 0$. Use the notation

$$\chi(A) = \begin{cases} 1 & \text{if property } A \text{ holds,} \\ 0 & \text{otherwise.} \end{cases}$$

The equilibrium equations are

$$(2.1) \quad [1 + \alpha\chi(i \geq 1) + \beta\chi(j \geq 1)]p_{ij} = \alpha p_{i+1,j} + \beta p_{i,j+1} + \chi(i \geq 1, j \geq 1)p_{i-1,j-1},$$

$i, j \geq 0.$

Multiply both sides of (2.1) by $z^i w^j$ and sum over i, j . We obtain the functional equation (1.1).

We prove the stability criterion mentioned in the introduction.

THEOREM 2.1. *The α, β -queue is stable iff $\alpha, \beta > 1$.*

Proof. From the theory of Markov chains [1, p. 54], it is known that $\lim_{t \rightarrow \infty} p_{ij}(t) = p_{ij}$ exists, for $i, j \geq 0$, and either:

- (i) $\forall p_{ij} > 0$ and $\sum p_{ij} = 1$ (the stable case), or
- (ii) $\forall p_{ij} = 0$ (the unstable case).

Let $P_i(t) = \sum_j p_{ij}(t)$, $Q_j(t) = \sum_i p_{ij}(t)$. $P_i(t)$ is the probability that there are i customers waiting or in service in the α -queue at time t , $Q_j(t)$ having a similar interpretation for the β -queue. The α -queue is stable iff $\alpha > 1$ [1, p. 77]. Hence for $\alpha \leq 1$, $\lim_{t \rightarrow \infty} P_i(t) = 0$, $0 \leq i < \infty$. Since $p_{ij}(t) \leq P_i(t)$, it follows that $\lim_{t \rightarrow \infty} p_{ij}(t) = 0$, $i, j \geq 0$, i.e., the α, β -queue is unstable. Similarly, $\beta \leq 1$ implies that the α, β -queue is unstable.

Suppose that $\alpha, \beta > 1$. Then $\lim_{t \rightarrow \infty} P_i(t) = P_i$, $\lim_{t \rightarrow \infty} Q_j(t) = Q_j$ with $\sum P_i = \sum Q_j = 1$. These conditions imply that we can choose $N, T \ni \sum_{i=N+1}^{\infty} P_i(t), \sum_{j=N+1}^{\infty} Q_j(t) \leq \frac{1}{4}$ for $t > T$. Hence

$$(2.2) \quad \sum_{0 \leq i, j \leq N} p_{ij}(t) \geq 1 - \sum_{i=N+1}^{\infty} P_i(t) - \sum_{j=N+1}^{\infty} Q_j(t) > \frac{1}{2}, \quad t > T.$$

Letting $t \rightarrow \infty$, we have $\sum_{1 \leq i, j \leq N} p_{ij} \geq \frac{1}{2}$, which rules out the unstable case. Hence the α, β -queue is stable for $\alpha, \beta > 1$.

The generating functions $P(z, 1) = \sum P_i z^i$, $Q(1, w) = \sum Q_j w^j$ of the α - and β -queues are obtained from (1.1) by letting $w = 1, z = 1$. We get $P(z, 1) = (\alpha - 1)/(\alpha - z)$, $P(1, w) = (\beta - 1)/(\beta - w)$, so that $P(z, 1), P(1, w)$ are rational functions. As will be shown in § 6, this is not the case for the generating function $P(z, w)$ of the α, β -queue.

We show here that $P(z, w)$ can be continued analytically beyond $|z|, |w| < 1$. This fact will be used in § 6.

THEOREM 2.2. $P(z, w) = \sum p_{ij} z^i w^j$ is analytic for $|z| < \sqrt{\alpha}, |w| < \sqrt{\beta}$.

Proof. From the formulas for $P(z, 1), P(1, w)$ we have

$$(2.3) \quad P_i = \frac{\alpha - 1}{\alpha^{i+1}}, \quad Q_i = \frac{\beta - 1}{\beta^{i+1}}, \quad 0 \leq i < \infty.$$

From $P_i = \sum_j p_{ij}$, $Q_j = \sum_i p_{ij}$, and (2.3), we have

$$(2.4) \quad p_{ij} \leq \min [P_i, Q_j] \leq \sqrt{P_i Q_j} \leq \frac{1}{\sqrt{\alpha \beta}}.$$

Equation (2.4) shows that $\sum p_{ij} z^i w^j$ converges for $|z| < \alpha, |w| < \beta$, which is the same as saying that it is analytic for these values of z, w .

3. The algebraic curve S . We describe the algebraic curve S and obtain for it a suitable parametrization. Our treatment is somewhat brief and the reader interested in details is referred to [7, Chap. 1]. Solving $Q = 0$ for either w or z , we get

$$(3.1) \quad w = \frac{(1 + \alpha + \beta)z - \alpha \pm \sqrt{D_1(z)}}{2z^2}, \quad z = \frac{(1 + \alpha + \beta)w - \beta \pm \sqrt{D_2(w)}}{2w^2},$$

where

$$(3.2) \quad D_1(z) = [(1 + \alpha + \beta)z - \alpha]^2 - 4\beta z^3, \quad D_2(w) = [(1 + \alpha + \beta)w - \beta]^2 - 4\alpha w^3.$$

Thus, S may be pictured as a two-sheeted Riemann surface over the z - or w -planes. We collect here various facts about the roots of the discriminants $D_1(z), D_2(w)$ needed in this paper. We assume from now on that $\beta \geq \alpha$.

THEOREM 3.1. (i) *The three zeros a_1, a_2, a_3 of $D_1(z)$ are real and distinct, with $0 < a_1 < a_2 < 1 < a_3$. $D_1(z) > 0$ on $(-\infty, a_1) \cup (a_2, a_3)$ and < 0 on $(a_1, a_2) \cup (a_3, \infty)$. A similar statement holds for the roots a'_1, a'_2, a'_3 , of $D_2(w)$.*

(ii) $a'_j = (\beta/\alpha)a_j$, $1 \leq j \leq 3$.

(iii) $\alpha \leq a_3 \leq \beta \leq a'_3$, any of the equality signs holding iff $\alpha = \beta$.

Proof. (i) We have

$$(3.3) \quad \begin{aligned} D_1(0) &= \alpha^2 > 0, & D_1\left(\frac{\alpha}{1+\alpha+\beta}\right) &= -4\beta\left(\frac{\alpha}{1+\alpha+\beta}\right)^3 < 0, \\ D_1(1) &= (\beta-1)^2 > 0, & D_1(+\infty) &= -\infty. \end{aligned}$$

It follows that $D_1(z)$ has roots a_1, a_2, a_3 contained respectively in the intervals $(0, \alpha/(1+\alpha+\beta))$, $(\alpha/(1+\alpha+\beta), 1)$, $(1, +\infty)$. The sign of $D_1(z)$ can then be determined via (3.3). A similar argument applies to $D_2(w)$.

(ii) Follows from $D_2(w) = (\beta^2/\alpha^2)D_1((\alpha/\beta)w)$.

(iii) We have

$$(3.4) \quad D_1(\alpha) = \alpha^2(\beta - \alpha)^2, \quad D_1(\beta) = [3\beta^2 + \beta + \alpha(\beta - 1)](\beta - 1)(\alpha - \beta),$$

$$(3.5) \quad D_2(\beta) = \beta^2(\beta - \alpha)^2.$$

Hence, for $\alpha < \beta$, $D_1(\beta) < 0 < D_1(\alpha)$ which implies $\alpha < a_3 < \beta$, and $D_2(\beta) > 0$ which implies $\beta < a'_3$. If $\alpha = \beta$, then $D_1(\beta) = 0$ so that $\beta = a_3 = a'_3$.

We now parametrize S . By the Schwarz–Christoffel transformation [6, Chapt. 5], $\exists$ a closed rectangle R_0 with vertices $0, \tilde{\omega}, \tilde{\omega} + \omega, \omega$, where $\tilde{\omega}, \omega/i > 0$, and a 1–1 mapping $t(z)$ from the extended closed upper half plane H to $R_0 \ni t(z)$ is continuous on H and analytic on $\text{Int } H = \text{interior of } H$. $t(z)$ maps a_1, a_2, a_3, ∞ to the four vertices in the order in which they are listed. Let $z = z(t)$ be the inverse mapping. By repeated use of the reflection principle, $z(t)$ can be extended as a meromorphic function on the entire t -plane. Furthermore $z(t)$ is an even and doubly periodic (=elliptic) function with period $2\tilde{\omega}, 2\omega$.

The function $w(t)$ is defined as follows. Let P be the z -plane minus the slits $[a_1, a_2], [a_3, \infty]$. For $z \in P$, let $\sqrt{D_1(z)}$ denote that square root which is analytic on P and $\sqrt{D_1(0)} = \alpha$. $z = z(t)$ maps each of strips $S_k = \{t | k \text{ Im } \omega < \text{Im } t < (k+1) \text{ Im } \omega\}$, k an integer, onto P . Let

$$(3.6) \quad s(t) = \begin{cases} \sqrt{D_1(z(t))}, & t \in S_k, \quad k \text{ even}, \\ -\sqrt{D_1(z(t))}, & t \in S_k, \quad k \text{ odd}. \end{cases}$$

For $\text{Im } t = k \text{ Im } \omega$, let $s(t) = \lim_{u \rightarrow t} s(u)$, $u \in S_k$. $s(t)$ is elliptic with periods $2\tilde{\omega}, 2\omega$ and we define

$$(3.7) \quad w(t) = \frac{(1 + \alpha + \beta)z(t) - \alpha + s(t)}{2z^2(t)}.$$

The mapping $t \rightarrow \phi(t) = (z(t), w(t))$ has the following properties: (i) $\phi(t)$ maps the t -plane onto S , (ii) $\phi(t)$ is locally 1–1; i.e., $\forall t \exists$ neighborhoods N_1 about t and N_2 about $\phi(t) \ni t \rightarrow \phi(t)$ is 1–1 from N_1 onto N_2 , (iii) $\phi(t_1) = \phi(t_2)$ iff $t_2 = t_1 + 2m\tilde{\omega} + 2n\omega$, m and n arbitrary integers. Thus S can be identified with the quotient group C/D , where C is the additive group of complex numbers and D the discrete subgroup $\{2m\tilde{\omega} + 2n\omega\}$, m and n integers. It follows that S is a torus, or a compact Riemann surface of genus 1.

We shall repeatedly use the following facts about elliptic functions [6, Chapt. 8].

THEOREM 3.2. Let $\omega_2 \neq 0$, ω_1/ω_2 not real. Let P be the fundamental parallelogram $\{t = r_1\omega_1 + r_2\omega_2 | 0 < r_1, r_2 \leq 1\}$ and $a_1, \dots, a_k, b_1, \dots, b_l \in P$. $\exists$ an elliptic function $f(t)$ with periods ω_1, ω_2 whose zeros and poles in P are given respectively by $\{a_1, \dots, a_k\}, \{b_1, \dots, b_l\}$ iff $k = l$ and $\sum a_i - \sum b_i = m_1\omega_1 + m_2\omega_2$, m_1 and m_2 integers,

in which case $f(t)$ is determined up to a constant multiple. $f(t)$ assumes every value k times in P , k being called the order of $f(t)$.

If the numbers $\alpha_1, \dots, \alpha_k$ satisfy $\alpha_i - a_i = m_{1i}\omega_1 + m_{2i}\omega_2$, m_{1i} and m_{2i} integers, then the α_i 's are called a complete set of zeros of $f(t)$. Similarly, we define a complete set of poles β_i . Theorem 3.2 implies that $\{\alpha_1, \dots, \alpha_k\}, \{\beta_1, \dots, \beta_k\}$ form complete sets of zeros and poles for $f(t)$ iff $\sum \alpha_i - \sum \beta_i = m_1\omega_1 + m_2\omega_2$, m_1 and m_2 integers.

Let R be the fundamental rectangle $\{t | -\tilde{\omega} < \operatorname{Re} t \leq \tilde{\omega}, -\operatorname{Im} \omega < \operatorname{Im} t \leq \operatorname{Im} \omega\}$. We obtain the values of $z(t)$, $w(t)$ at certain points of R , as these will be required later on. For any two complex numbers a, b , the symbols $[a, b]$, (a, b) , $[a, b)$, $(a, b]$ denote respectively the closed, open, and half open line segments directed from a to b , $[a, b)$ containing a and $(a, b]$ containing b . $z(t)$ maps $[\omega, 0]$ 1-1 onto $[-\infty, a_1]$, and $[\tilde{\omega}, \tilde{\omega} + \omega]$ 1-1 onto $[a_2, a_3]$. As $0 < a_1, a_2 < 1 < \alpha \leq a_3$, $\exists$ unique points $m \in (0, \omega)$, $p, q \in (\tilde{\omega}, \tilde{\omega} + \omega) \ni z(m) = 0$, $z(p) = 1$, $z(q) = \alpha$. Furthermore p precedes q on $(\tilde{\omega}, \tilde{\omega} + \omega]$, which we denote by $p < q$.

THEOREM 3.3. (i) The zeros and poles of $z(t)$ in R are respectively $\{m, -m\}$, $\{\omega, \omega\}$ (i.e., ω is a double pole of $z(t)$).

(ii) The zeros and poles of $w(t)$ in R are respectively $\{m, \omega\}, \{-m, -m\}$.

(iii) $m = \omega/3$.

(iv) $w(t) = (\beta/\alpha)z(t-2m)$.

(v) $(p+q)/2 = \tilde{\omega} + 2m$. Hence

$$(3.8) \quad \tilde{\omega} + m \leq p < \tilde{\omega} + 2m < q \leq \tilde{\omega} + \omega,$$

with either equality holding iff $\alpha = \beta$.

(vi) At the points $p-2km, q-2km, k=0, 1, 2$, $z(t)$ and $w(t)$ assume the values listed in Table 1. The positions of the points p, q, m in R are indicated in Fig. 1.

TABLE 1

t	$p-4m$	$q-4m$	$p-2m$	$q-2m$	p	q
$z(t)$	α	1	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta}$	1	α
$w(t)$	$\frac{\beta}{\alpha}$	β	β	$\frac{\beta}{\alpha}$	1	1

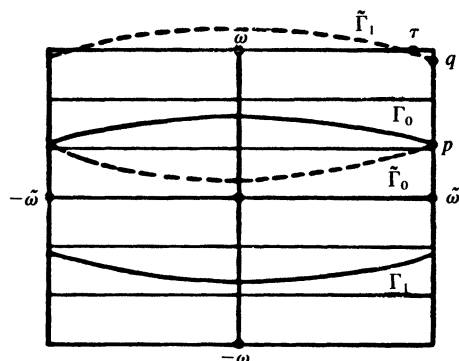


FIG. 1

Proof. (i) Follows from the mapping properties of $z(t)$.

(ii) Equation (3.7) shows that the zeros and poles of $w(t)$ are among those of $z(t)$. Choose $\sqrt{D_1(z)}$ as above. For z near 0, we have

$$(3.9) \quad \begin{aligned} w_1(z) &= \frac{(1+\alpha+\beta)z - \alpha + \sqrt{D_1(z)}}{2z^2} = -\frac{\beta}{\alpha}z + \dots, \\ w_2(z) &= \frac{(1+\alpha+\beta)z - \alpha - \sqrt{D_1(z)}}{2z^2} = -\frac{\alpha}{z^2} + \dots. \end{aligned}$$

For t near m , $w(t) = w_1(z(t))$ and for t near $-m$, $w(t) = w_2(z(t))$. It follows from (3.9) that $w(t)$ has a simple zero at m and a double pole at $-m$. Equation (3.7) shows that $w(t)$ has a simple zero at ω .

(iii) Applying Theorem 3.2 to $w(t)$, we have $\omega + 3m + 2j\tilde{\omega} + 2k\omega = 0$, for some integers j and k . Let $m = c\omega$, $0 < c < 1$. Then

$$2j\tilde{\omega} + (2k+1+3c)\omega = 0, \quad \text{so that } j=0, \quad k=-1, \quad c=\frac{1}{3}.$$

Hence $m = \omega/3$.

(iv) From (i)–(iii), we conclude that $z(t-2m)$, $w(t)$ have identical sets of zeros and poles. Hence $w(t) = cz(t-2m)$, c a constant. To evaluate c , let $t = m + \tau$ and $\tau \rightarrow 0$. $z(t)$ has a simple zero at m , so that

$$(3.10) \quad w(m+\tau) = cz(m-\tau) \sim -cz(m+\tau) \quad \text{as } \tau \rightarrow 0.$$

Equation (3.9) gives

$$(3.11) \quad w(m+\tau) \sim -\frac{\beta}{\alpha}z(m+\tau) \quad \text{as } \tau \rightarrow 0.$$

Comparing (3.10) and (3.11), we get $c = \beta/\alpha$.

(v) Letting $t = p, q$ in (3.7), we get $w(p) = w(q) = 1$. Applying the reasoning used in (iii) to $w(t) - 1$, we obtain $(p+q)/2 = \tilde{w} + 2m$. Observe that $q = \tilde{\omega} + \omega$ iff $\alpha = a_3$ which, by Theorem 3.1(iii), holds iff $\alpha = \beta$.

(vi) By the reflection principle $z(t) = \overline{z(\bar{t})}$, so that $z(\bar{p}) = 1$, $z(\bar{q}) = \alpha$. Letting $t = \bar{p}, \bar{q}$ in (3.7), we get $w(\bar{p}) = \beta$, $w(\bar{q}) = \beta/\alpha$. By (v), $\bar{p} = q - 4m$, $\bar{q} = p - 4m$. We have thus obtained the values of $z(t)$, $w(t)$ at $p, q, p-4m, q-4m$. The values of $z(t)$, $w(t)$ at $p-2m, q-2m$ are then obtained from (iv) letting $t = p, q, p-2m, q-2m$.

4. The regions D_z, D_w . Let $D_z = \{t \mid |z(t)| < 1\}$, $D_w = \{t \mid |w(t)| < 1\}$. The closures of these sets are $\bar{D}_z = \{t \mid |z(t)| \leq 1\}$, $\bar{D}_w = \{t \mid |w(t)| \leq 1\}$. The functions $P(z(t), 0)$, $P(0, w(t))$ are analytic respectively on D_z, D_w and continuous on $\bar{D}_z, \bar{D}_w$. Let D_z, D_w be the components of D_z, D_w containing 0 and $2m$, respectively. In § 5, we show that $P(z(t), 0)$, $P(0, w(t))$ can be continued analytically from D_z, D_w to the entire t -plane. We digress here to give a detailed description of D_z, D_w and how they interlace, as this information will be required in later sections.

From the mapping properties of $z(t)$ and $w(t) = (\beta/\alpha)z(t-2m)$, it follows that both D_z, D_w consist of an infinite number of simply connected components. The component D_z is a horizontal curvilinear strip contained in the horizontal rectangular strip $\{t \mid |\operatorname{Im} t| < \operatorname{Im} \omega\}$. The other components of D_z are the translates of D_z by $2k\omega$, k any integer $\neq 0$. Similarly, D_w is a horizontal curvilinear strip contained in the horizontal rectangular strip $\{t \mid |\operatorname{Im}(t-2m)| < \operatorname{Im} \omega\}$ and the other components of D_w are the translates of D_w by $2k\omega$, k any integer $\neq 0$.

Let Γ_1, Γ_0 be the lower and upper boundaries of D_z , and $\tilde{\Gamma}_0, \tilde{\Gamma}_1$ the lower and upper boundaries of D_w . The mapping properties of $z(t)$, $w(t)$ imply that: D_z, D_w are

invariant under $t \rightarrow t + 2\omega$, D_z is symmetric with respect to the real and imaginary axes, D_w is symmetric with respect to the horizontal line through $2m$ and imaginary axis. Thus a complete description of the curves $\Gamma_j, \tilde{\Gamma}_j$ ($j=0, 1$) is obtained by describing their portions in the vertical strip $\{t | 0 \leq \operatorname{Re} t \leq \tilde{\omega}\}$. This is done in Theorem 4.1, whose contents are illustrated in Fig. 1.

THEOREM 4.1. (i) $\Gamma_0 \subset \{t | \operatorname{Im} m \leq \operatorname{Im} t < 2 \operatorname{Im} m\}$. Γ_0 touches the line $\operatorname{Im} t = \operatorname{Im} m$ iff $\alpha = \beta$ and then only at $t = \tilde{\omega} + m$.

(ii) $\tilde{\Gamma}_1 \subset \{t | 2 \operatorname{Im} m < \operatorname{Im} t < 4 \operatorname{Im} m\}$. Let τ be the point on $(\omega, \tilde{\omega} + \omega]$ where $z(\tau) = \beta$. $\tilde{\Gamma}_1$ crosses $[\omega, \tilde{\omega} + \omega]$ at τ with $[\omega, \tau]$ lying below and $(\tau, \tilde{\omega} + \omega]$ above $\tilde{\Gamma}_1$.

(iii) $\Gamma_0 \cap \tilde{\Gamma}_0 = p$, p defined as in Theorem 3.3. Except for p , all points of Γ_0 are in D_w and all points of $\tilde{\Gamma}_0$ are in D_z .

Proof. (i) $w(t) = (\beta/\alpha)z(t-2m)$ maps $[2m, \tilde{\omega} + 2m]$ 1-1 onto $[a'_1, a'_2]$, and $[\tilde{\omega} - m, -m]$ 1-1 onto $[a'_3, \infty]$. The discriminant $D_2(w) < 0$ on $(a'_1, a'_2) \cup (a'_3, \infty)$. It follows that for $t \in [2m, \tilde{\omega} + 2m] \cup [\tilde{\omega} - m, -m]$, the roots of $Q(z, w(t)) = 0$ are complex conjugates $z(t), \bar{z}(t)$ whose product $|z(t)|^2 = \alpha/w(t)$.

By Theorem 3.1, $a'_2 < \alpha \leq a'_3$, equality holding iff $\alpha = \beta$. Hence $|z(t)| > 1$, $t \in [2m, \tilde{\omega} + 2m]$. Let R_0 be the rectangle of § 3. $|z(t)| > 1$ on the part of R_0 above Γ_0 and < 1 on the part below Γ_0 . Hence Γ_0 lies below the line $\operatorname{Im} t = 2 \operatorname{Im} m$. Similarly, $|z(t)| \leq 1$ on $[\tilde{\omega} - m, -m]$, equality occurring only for $\alpha = \beta$ and $t = \tilde{\omega} - m$. Hence for $\alpha < \beta$, Γ_1 lies below $\operatorname{Im} t = -\operatorname{Im} m$. By symmetry this is equivalent to Γ_0 lying above $\operatorname{Im} t = \operatorname{Im} m$. If $\alpha = \beta$, we likewise conclude that Γ_0 lies above $\operatorname{Im} t = \operatorname{Im} m$, except at the end point $\omega + m$ where Γ_0 touches this line.

(ii) From the mapping properties of $w(t)$, $\tilde{\Gamma}_1$ lies above $\operatorname{Im} t = 2 \operatorname{Im} m$. The rest of the reasoning is analogous to that of (i). For $t \in [0, \tilde{\omega}] \cup [\tilde{\omega} + \omega, \omega]$, $|w(t)|^2 = \beta/z(t)$. Since $z(t)$ maps $[0, \tilde{\omega}]$ 1-1 onto $[a_1, a_2]$ and $a_2 < \beta$, we conclude that $|w(t)| > 1$ on $[0, \tilde{\omega}]$. Hence $\tilde{\Gamma}_0$ lies above $\operatorname{Im} t = 0$ which, by symmetry, is equivalent to $\tilde{\Gamma}_1$ lying below $\operatorname{Im} t = 4 \operatorname{Im} m$. As $z(t)$ maps $[\tilde{\omega} + \omega, \omega]$ 1-1 onto $[a_3, \infty]$ and $a_3 \leq \beta$, we conclude that $|w(t)| < 1$ on $[\omega, \tau]$ and > 1 on $(\tau, \tilde{\omega} + \omega]$, i.e., $[\omega, \tau]$ lies below and $(\tau, \tilde{\omega} + \omega]$ above $\tilde{\Gamma}_1$.

(iii) Let $z = z(t)$, $w = w(t)$, $t \in \Gamma_0 \cap \tilde{\Gamma}_0$. $Q(z, w) = 0$ gives

$$(4.1) \quad |(zw)^2 + \alpha w + \beta z| = 1 + \alpha + \beta.$$

Since $|z| = |w| = 1$, (4.1) holds iff $(zw)^2 = w = z \Leftrightarrow z = w = 1, \zeta, \zeta^2$ where $\zeta = e^{2\pi i/3}$. The solutions $z = w = \zeta, \zeta^2$ do not satisfy $Q(z, w) = 0$. Hence $z(t) = w(t) = 1$, i.e., $t = p$.

Let $\Gamma'_0 = \Gamma_0 - \{p\}$, $\tilde{\Gamma}'_0 = \tilde{\Gamma}_0 - \{p\}$. From what we have shown so far, Γ'_0 penetrates D_w and does not meet its boundary curves. Hence $\Gamma'_0 \subset D_w$. Analogously we reason that $\tilde{\Gamma}'_0 \subset D_z$.

5. Analytic continuation of $P(z(t), 0)$, $P(0, w(t))$. Let $A(t) = P(z(t), 0)$, $t \in \bar{D}_z$, and $B(t) = P(0, w(t))$, $t \in \bar{D}_w$. $A(t)$, $B(t)$ are respectively analytic on D_z , D_w and continuous on $\bar{D}_z$, $\bar{D}_w$.

THEOREM 5.1. $A(t)$, $B(t)$ can be continued as meromorphic functions to the entire t -plane. The extended functions satisfy

$$(5.1) \quad A(-t) = A(t), \quad A(t + 2\tilde{\omega}) = A(t), \quad A(t - 4m) = e(t)A(t),$$

$$(5.2) \quad B(t) = R(t)A(t)$$

with

$$(5.3) \quad f(z) = \frac{z-1}{z}, \quad R(t) = -\frac{\beta}{\alpha} \frac{f(w(t))}{f(z(t))}, \quad e(t) = \frac{f(z(t-4m))}{f(z(t))}.$$

Proof. $\bar{D}_z$ is invariant under $t \rightarrow -t$, $t \rightarrow t + 2\tilde{\omega}$, and $z(t) = z(-t) = z(t + 2\omega)$. Hence

$$(5.4) \quad A(t) = A(-t) = A(t + 2\tilde{\omega}), \quad t \in \bar{D}_z.$$

The fact that $\bar{D}_w$ is invariant under $t \rightarrow 4m - t$, $t \rightarrow t + 2\tilde{\omega}$, and $w(t) = (\beta/\alpha)z(t - 2m)$ implies $w(t) = w(4m - t) = w(t + 2\tilde{\omega})$. Hence

$$(5.5) \quad B(t) = B(4m - t) = B(t + 2\tilde{\omega}), \quad t \in \bar{D}_w.$$

Letting $z = z(t)$, $w = w(t)$ in (1.1) we get

$$(5.6) \quad R(t)A(t) = B(t), \quad t \in \bar{D}_z \cap \bar{D}_w.$$

Because of (5.6), we may continue $A(t)$ to $\bar{D} = \bar{D}_z \cup \bar{D}_w$ by defining $A(t) = B(t)/R(t)$, $t \in \bar{D}_w$. Let $\bar{D} + 4m$ be the translate of $\bar{D}$ by $4m$, and $\bar{D}^*$ the image of $\bar{D}_z \cap \bar{D}_w$ under $t \rightarrow 4m - t$. Observe that $\bar{D}^* = \bar{D} \cap [\bar{D} + 4m]$. Let $t \in \bar{D}^*$. As $t \in \bar{D}_w$ and $4m - t \in \bar{D}_z \cap \bar{D}_w$, we conclude from (5.4)–(5.6) that

$$(5.7) \quad R(t)A(t) = B(t) = B(4m - t) = R(4m - t)A(4m - t) = R(4m - t)A(t - 4m),$$

i.e.,

$$(5.8) \quad A(t - 4m) = \frac{R(t)}{R(4m - t)} A(t) = e(t)A(t), \quad t \in \bar{D}^*.$$

Because of (5.8), we may continue $A(t)$ to $\bar{D} + 4m$ by defining $A(t) = A(t - 4m)/e(t)$. Repeating this procedure, we continue $A(t)$ as a meromorphic function to $\bigcup_{k=-\infty}^{\infty} [\bar{D} + 4mk] = t$ -plane. The continuation of $B(t)$ is defined by $B(t) = R(t)A(t)$. Finally, each of the relations in (5.1) has been shown to hold on a portion of the t -plane. They remain valid for all t by analytic continuation.

6. Determination of $P(z, w)$. We determine $P(z, 0)$, $P(0, w)$, and hence $P(z, w)$. In the formulas (6.1), (6.2) below the constants $\sqrt{a_3 - 1}$, etc. are assumed positive and the functions $\sqrt{a_3 - z}$, $\sqrt{a'_3 - w}$ are those analytic branches which are positive at the origin.

THEOREM 6.1. *Let*

$$(6.1) \quad \phi(z) = \frac{\sqrt{a_3 - z} + \sqrt{a_3 - 1}}{[\sqrt{a_3 - z} + \sqrt{a_3 - \alpha/\beta}] \cdot [\sqrt{a_3 - z} - \sqrt{a_3 - \alpha}]}, \quad |z| \leq 1,$$

$$(6.2) \quad \psi(w) = \frac{\sqrt{a'_3 - w} + \sqrt{a'_3 - 1}}{[\sqrt{a'_3 - w} + \sqrt{a'_3 - \beta/\alpha}] \cdot [\sqrt{a'_3 - w} - \sqrt{a'_3 - \beta}]}, \quad |w| \leq 1.$$

Then

$$(6.3) \quad P(z, 0) = \frac{\beta - 1}{\beta} \frac{\phi(z)}{\phi(1)}, \quad |z| \leq 1,$$

$$(6.4) \quad P(0, w) = \frac{\alpha - 1}{\alpha} \frac{\psi(w)}{\psi(1)}, \quad |w| \leq 1.$$

For $\alpha = \beta$, (6.3), (6.4) become

$$(6.5) \quad P(z, 0) = P(0, z) = \frac{(\alpha - 1)^{3/2}}{\alpha(\alpha - z)^{1/2}}.$$

We remark that these formulas provide analytic continuations of $P(z, 0)$, $P(0, w)$ to the z - and w -planes minus the respective slits $[a_3, \infty)$, $[a'_3, \infty)$. If $\alpha < \beta$, then $P(z, 0)$ is analytic in the slit z -plane except for a simple pole at α , while $P(0, w)$ is analytic

in the slit w -plane. If $\alpha = \beta$, then both $P(z, 0)$, $P(0, w)$ are analytic in the slit z - and w -planes.

Proof. We first determine $A(t)$, $B(t)$ by showing that they are elliptic functions with periods $2\tilde{\omega}$, 4ω , and by obtaining their zeros and poles. Using this information, we identify $A(t)$, $B(t)$ respectively as rational functions of $\sqrt{a_3 - z(t)}$, $\sqrt{a'_3 - w(t)}$ (the determination of the square roots are made precise below). The formulas for $P(z, 0)$, $P(0, w)$ follow directly from those for $A(t)$, $B(t)$.

Let $A_1(t) = A(t)/f(z(t))$, where $f(z) = (z - 1)/z$. Equations (5.1) may be rewritten as

$$(6.6) \quad A_1(-t) = A_1(t), \quad A_1(t + 2\tilde{\omega}) = A_1(t), \quad A_1(t + 4m) = A_1(t).$$

Equation (6.6) states that $A_1(t)$ is an even elliptic function with periods $2\tilde{\omega}$, $4m$. Hence $A(t)$ is an even elliptic function with periods $2\tilde{\omega}$, 4ω . We locate the zeros and poles of $A_1(t)$ in the fundamental rectangle $R_1 = \{t | -\tilde{\omega} < \operatorname{Re} t \leq \tilde{\omega}, -2 \operatorname{Im} m < \operatorname{Im} t \leq 2 \operatorname{Im} m\}$. We first show that $A(t)$ is analytic in the strip $T_1 = \{t | |\operatorname{Im} t| \leq 2 \operatorname{Im} m\}$.

Using Theorem 3.1, the zeros of $R(t)$ in R are computed to be q , $-m$, which are outside D_w . It follows that $A(t)$ is analytic in $D^0 = \text{interior of } \bar{D}$. Let $T_2 = \{t | 0 \leq \operatorname{Im} t \leq 2 \operatorname{Im} m\}$. $T_2 \subset D^0$, so that $A(t)$ is analytic in T_2 . Since $A(-t) = A(t)$, we conclude that $A(t)$ is analytic in T_1 .

Again by Theorem 3.1, the zeros and poles of $f(z(t))$ in either R or R_1 are computed to be $q - 4m$, p and $\pm m$. Since $A(q - 4m) = A(p) = P(1, 0) = (\beta - 1)/\beta > 0$, we conclude that $q - 4m$, p are the poles of $A_1(t)$ in R_1 . $A_1(t)$ has zeros at $\pm m$. Since the number of zeros in R_1 must equal the number of poles, $\pm m$ are the only zeros of $A_1(t)$ in R_1 .

Knowledge of the zeros and poles of $A_1(t)$ leads at once to the following complete sets of zeros and poles for $A(t)$:

$$\{-\omega, \omega, q + 2m, p + 6m\} \text{ and } \{p - 4m, q, p + 4m, q + 4m\}, \quad \alpha < \beta.$$

For $\alpha = \beta$, we have $q + 2m = p + 4m$, $p + 6m = q + 4m$, and the above complete sets are to be replaced by: $\{\pm\omega\}$ and $\{p - 4m, q\}$.

Let $\lambda(t)$ be an elliptic function with periods $2\tilde{\omega}$, 4ω , having the complete sets of zeros and poles $\{\tilde{\omega} \pm \omega\}$ and $\{\pm\omega\}$. The existence of $\lambda(t)$ follows from Theorem 3.2, as in this case $\sum a_i - \sum b_i = 2\tilde{\omega}$. $\lambda^2(t)$ and $a_3 - z(t)$ have identical zeros and poles. Multiplying by a constant, we may assume that $\lambda(t) = \sqrt{a_3 - z(t)}$, where $\sqrt{a_3 - z(t)} \geq 0$ on $[\tilde{\omega} - \omega, \tilde{\omega} + \omega]$, and ≤ 0 on $[\tilde{\omega} + \omega, \tilde{\omega} + 3\omega]$. Now $p - 4m, q \in [\tilde{\omega} - \omega, \tilde{\omega} + \omega]$ and $q + 2m, p + 4m, q + 4m, p + 6m \in [\tilde{\omega} + \omega, \tilde{\omega} + 3\omega]$. We conclude from Theorem 3.3(vi) that

$$(6.7) \quad \begin{aligned} \lambda(p - 4m) &= \lambda(q) = \sqrt{a_3 - \alpha}, \quad \lambda(q + 2m) = \lambda(p + 6m) = -\sqrt{a_3 - 1}, \\ \lambda(p + 4m) &= \lambda(q + 4m) = -\sqrt{a_3 - \frac{\alpha}{\beta}}, \end{aligned}$$

each of the square roots ≥ 0 . It follows from (6.7) that $A(t)$ and

$$(6.8) \quad F(t) = \frac{\lambda(t) + \sqrt{a_3 - 1}}{(\lambda(t) + \sqrt{a_3 - (\alpha/\beta)})(\lambda(t) - \sqrt{a_3 - 1})}$$

have identical zeros and poles, so that $A(t) = cF(t)$ for some constant c .

Let t vary over $\bar{D}_z$. Then $z(t)$ varies over $|z| \leq 1$. Furthermore $A(t) = P(z(t), 0)$ and $\lambda(t) = \sqrt{a_3 - z(t)}$, where $\sqrt{a_3 - z}$ is defined as in Theorem 6.1. We conclude from (6.8) that $P(z, 0) = c\phi(z)$, $|z| \leq 1$, where $\phi(z)$ is given by (6.1). To evaluate c , let $z = 1$. We have $c\phi(1) = P(1, 0) = (\beta - 1)/\beta$, and so (6.3) holds.

Determination of $B(t)$ proceeds analogously. Complete sets of zeros and poles of $B(t) = R(t)A(t)$ are given by:

$$\{-m, 5m, p+6m, q+6m\} \text{ and } \{p+4m, q+4m, p+8m, q+8m\} \quad \text{if } \alpha < \beta, \\ \{-m, 5m\} \text{ and } \{p+4m, q+8m\} \quad \text{if } \alpha = \beta.$$

Hence

$$(6.9) \quad B(t) = c' \frac{\mu(t) + \sqrt{a'_3 - 1}}{(\mu(t) + \sqrt{a'_3 - (\beta/\alpha)}) \cdot (\mu(t) + \sqrt{a'_3 - \beta})},$$

where c' is a constant and $\mu(t) = \sqrt{a'_3 - w(t)}$, $\mu(t) \geq 0$ on $[\tilde{w} - m, \tilde{w} + 5m]$ and ≤ 0 on $[\tilde{w} + 5m, \tilde{w} + 11m]$. Equation (6.9) yields (6.4).

From the formulas in Theorem 6.1, we can compute various quantities of interest for the double queue. We compute the correlation coefficient $\rho(X_1, X_2)$.

THEOREM 6.2. For $\alpha = \beta$

$$(6.10) \quad p(X_1, X_2) = \frac{1}{2} - \frac{1}{8\alpha}.$$

We remark that the method of derivation also yields a formula for ρ when $\alpha < \beta$. The formula is however cumbersome in this case and we omit it.

Proof. For $\alpha = \beta$, X_1 and X_2 are identically distributed so that

$$(6.11) \quad \rho(X_1, X_2) = \frac{E(X_1 X_2) - E^2(X_1)}{E(X_1^2) - E^2(X_1)}.$$

From $P(z, 1) = (\alpha - 1)/(\alpha - z)$, one readily obtains

$$(6.12) \quad E(X_1) = \frac{1}{\alpha - 1}, \quad E(X_1^2) = \frac{2}{(\alpha - 1)^2} + \frac{1}{\alpha - 1}.$$

To compute $E(X_1, X_2)$ expand both sides of (1.1) into Taylor series about (1.1), which is permissible by Theorem 2.2, and equate coefficients of $(z - 1)^2(w - 1)$. We obtain

$$(6.13) \quad E(X_1, X_2) = P_{zw}(1, 1) = \frac{1}{2(\alpha - 1)} + \frac{11}{8} \frac{1}{(\alpha - 1)^2},$$

(6.10) follows from (6.11)–(6.13).

7. Asymptotic formula for p_{i0}, p_{0j} . The formulas for $P(z, 0), P(0, w)$ of § 5 can be used to obtain the asymptotic behavior of p_{i0}, p_{0j} as $i \rightarrow \infty, j \rightarrow \infty$.

THEOREM 7.1. (i) For $\alpha < \beta$

$$(7.1) \quad p_{i0} \sim \frac{c_1}{\alpha^i} \quad \text{as } i \rightarrow \infty.$$

(ii) For $\alpha < \beta$

$$(7.2) \quad p_{0j} \sim \frac{c_2}{j^{3/2} \alpha_j^{1/2}} \quad \text{as } j \rightarrow \infty.$$

(iii) For $\alpha = \beta$,

$$(7.3) \quad p_{i0} = p_{0i} \sim \frac{c_3}{i^{1/2} \alpha^i} \quad \text{as } i \rightarrow \infty,$$

where

$$(7.4) \quad \begin{aligned} a &= \sqrt{a_3 - \alpha}, \quad b = \sqrt{a_3 - 1}, \quad c = \sqrt{a_3 - \left(\frac{\alpha}{\beta}\right)}, \quad d = \sqrt{a_3}, \quad c_1 = \frac{a-1}{\alpha} \frac{\beta-\alpha}{\beta} \\ c_2 &= \frac{1}{4\sqrt{\pi}} \frac{(\alpha-1)(a+c)(b+c)d\left(\frac{1}{a} + \frac{1}{b} - \frac{1}{c}\right)}{\alpha ab}, \quad c_3 = \sqrt{\frac{2}{\pi}} \left(\frac{\alpha-1}{\alpha}\right)^{3/2} \end{aligned}$$

Proof. (i) From (6.4), (6.5), we have

$$(7.5) \quad P(z, 0) = \frac{r}{z - \alpha} + g(z), \quad |z| < a_3,$$

where r is the residue of $P(z, 0)$ at α and $g(z)$ is analytic for $|z| < a_3$. Hence

$$(7.6) \quad p_{i0} \sim \frac{-r}{\alpha^{i+1}} \quad \text{as } i \rightarrow \infty.$$

We evaluate r as follows. Since $z(q) = \alpha$, $w(q) = 1$, we have

$$(7.7) \quad \begin{aligned} R(t) &\sim \frac{-\beta}{\alpha - 1} w'(q)(t - q), \quad A(t) = P(z(t), 0) \sim \frac{r}{z'(q)} \frac{1}{t - q} \quad \text{as } t \rightarrow q, \\ B(q) &= P(0, w(q)) = P(0, 1) = \frac{\alpha - 1}{\alpha}. \end{aligned}$$

Hence

$$(7.8) \quad \lim_{t \rightarrow q} R(t)A(t) = \frac{-\beta r}{\alpha - 1} \frac{w'(q)}{z'(q)} = \frac{\beta r}{\alpha - 1} \cdot \frac{(\partial Q / \partial z)(\alpha, 1)}{(\partial Q / \partial w)(\alpha, 1)} = -\frac{\beta r}{\alpha(\beta - \alpha)}.$$

Since $R(t)A(t) = B(t)$, we conclude from (7.7), (7.8) that

$$(7.9) \quad r = -\frac{(\alpha - 1)(\beta - \alpha)}{\beta}.$$

Equations (7.6), (7.9) give (7.1).

(ii) We first prove the following lemma:

LEMMA 7.1. Let $0 < \varepsilon < 1$ and $\sqrt{1 - z}$ analytic in $|z| < 1$ with $\sqrt{1 - z} = 1$ at $z = 0$.

Let

$$\frac{1}{\varepsilon + \sqrt{1 - z}} = \sum_{n=0}^{\infty} a_n z^n, \quad |z| < 1.$$

Then

$$(7.10) \quad a_n \sim \frac{1}{2\sqrt{\pi}\varepsilon^2 n^{3/2}} \quad \text{as } n \rightarrow \infty.$$

Proof. We have

$$(7.11) \quad \sqrt{1 - z} = 1 - \sum_{n=1}^{\infty} b_n z^n, \quad |z| < 1,$$

where

$$(7.12) \quad b_n = \binom{2n-2}{n-1} \frac{1}{n 2^{2n-1}} > 0, \quad 1 \leq n < \infty, \quad b_n \sim \frac{1}{2\sqrt{\pi}} \frac{1}{n^{3/2}} \quad \text{as } n \rightarrow \infty,$$

by Stirling's formula.

Write

$$(7.13) \quad \frac{1}{\varepsilon + \sqrt{1-z}} = \frac{\sqrt{1-z-\varepsilon}}{(1-\varepsilon^2)-z} \\ = \left[(1-\varepsilon) - \sum_{n=1}^{\infty} b_n z^n \right] \sum_{n=0}^{\infty} \frac{z^n}{(1-\varepsilon^2)^{n+1}}, \quad |z| < 1-\varepsilon^2,$$

so that

$$(7.14) \quad a_n = \frac{1-\varepsilon - \sum_{k=1}^n b_k (1-\varepsilon^2)^k}{(1-\varepsilon^2)^{n+1}}, \quad n \geq 1.$$

Setting $z = 1-\varepsilon^2$ in (7.11) gives

$$(7.15) \quad \sum_{k=1}^{\infty} b_k (1-\varepsilon^2)^k = 1-\varepsilon.$$

We conclude from (7.14), (7.15) that

$$(7.16) \quad a_n = b_{n+1} \sum_{k=0}^{\infty} \frac{b_{n+1+k}}{b_{n+1}} (1-\varepsilon^2)^k, \quad n \geq 1.$$

$(b_{n+1+k}/b_{n+1}) \leq 1$, $n, k \geq 0$, and $\lim_{n \rightarrow \infty} (b_{n+1+k}/b_{n+1}) = 1$, $k \geq 0$. Hence by the dominated convergence theorem, the series in (7.16) $\rightarrow \sum_{k=0}^{\infty} (1-\varepsilon^2)^k = 1/\varepsilon^2$ as $n \rightarrow \infty$. Equation (7.10) follows from (7.12), (7.16).

We now derive (7.2). Write

$$(7.17) \quad \psi(w) = \frac{A_1}{\sqrt{1-w/a'_3} + \sqrt{1-\beta/\alpha a'_3}} + \frac{A_2}{\sqrt{1-w/a'_3} + \sqrt{1-\beta/a'_3}},$$

where A_1, A_2 are constants. Applying the lemma with $z = w/a'_3$ and evaluating A_1, A_2 , we obtain (7.2)

(iii) For $\alpha = \beta$, the formulas for ϕ, ψ simplify to

$$(7.18) \quad \phi(z) = \psi(z) = \frac{1}{\sqrt{\alpha-z}} = \frac{1}{\sqrt{\alpha}} \sum_{n=0}^{\infty} \frac{1}{2^{2n}} \binom{2n}{n} \frac{z^n}{\alpha^n}.$$

Equation (7.3) follows from (6.5), (7.18), after applying Stirling's formula.

Theorem 7.1 gives the asymptotic behavior of p_{ij} along the i - and j -axes. This result can be generalized to all lines parallel to either of the axes.

THEOREM 7.2. *Let c_1, c_2, c_3 be defined as in Theorem 7.1.*

(i) *If $\alpha < \beta$, then*

$$(7.19) \quad p_{ij} \sim \frac{c_1}{\alpha^{i-j}\beta^j} \quad \text{for fixed } j \text{ and } i \rightarrow \infty.$$

(ii) *If $\alpha < \beta$, then*

$$(7.20) \quad p_{ij} \sim c_2 \frac{1 + [1 - (\alpha/a'_3)^{1/2}]i}{j^{3/2} \alpha^{1/2} (a'_3)^{j-i/2}} \quad \text{for fixed } i \text{ and } j \rightarrow \infty.$$

(iii) *If $\alpha = \beta$, then*

$$(7.21) \quad p_{ij} \sim \frac{c_3}{i^{1/2} \alpha^i} \quad \text{for fixed } j \text{ and } i \rightarrow \infty.$$

Theorem 7.2 follows readily from Theorem 7.1 and the equilibrium equations (2.1), using a standard induction argument on i and j . The details of the proof are omitted.

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